NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

AIRCRAFT AIRFRAME TUBING, CARBON AND ALLOY STEEL SEAMLESS ROUND

AIRFRAME TUBING IS COMMONLY PRODUCED STRESS RELIEVED, ANNEALED, NORMALIZED OR OTHERWISE HEAT TREATED TO OBTAIN PROPERTIES REQUIRED FOR AIRBORNE STRUCTURAL PARTS, AND IS GENERALLY USED IN THE CONDITION RECEIVED FROM THE MILL. IT IS NOT INTENDED FOR AIRCRAFT PARTS WHERE MACHINING OUTSIDE OR INSIDE SURFACES TO EXACT DIMENSIONS IS NECESSARY.

CROSS SECTIONAL DIMENSIONAL AND WALL THICKNESS TOLERANCES:

NOMINAL OD, INCHES	(a) (c) OD TOLERANCE, INCH, PLUS OR MINUS		(b)
	ANNEALED, NORMALIZED OR COLD DRAWN AND STRESS RELIEVED	QUENCHED AND TEMPERED	WALL THICKNESS TOLERANCE % + OR -
3/16 TO (NOT INCLUDING) 1/2	.004	.010	15
1/2 TO (NOT INCLUDING) 1-1/2	.005	.015	10
1-1/2 TO (NOT INCLUDING) 3-1/2	.010	.030	10
3-1/2 TO (INCLUDING) 5-1/2	.015	.045	10

(a) TOLERANCES ARE APPLICABLE TO INDIVIDUAL MEASUREMENTS.

(b) FOR TUBING WITH INSIDE DIAMETER LESS THAN 1/2 INCH (OR LESS THAN 5/8 INCH WHEN THE WALL THICKNESS IS MORE THAN 20% OF THE OUTSIDE DIAMETER) WHICH CANNOT BE SUCCESSFULLY DRAWN OVER A STATIONARY MANDREL, THE WALL THICKNESS MAY VARY 15% OVER OR UNDER THAT SPECIFIED.

(c) TUBING WITH WALL THICKNESS LESS THAN 3% OF THE NOMINAL OD WILL BE ALLOWED AN OVALITY TOLERANCE OF ±1/2% OF THE NOMINAL OD IN ADDITION TO THE TOLERANCE INDICATED.

STRAIGHTNESS:

(1) AS FINISHED, ANNEALED, NORMALIZED OR STRESS RELIEVED.

NOMINAL OD INCHES	WALL THICKNESS	MAXIMUM CURVATURE (DEPTH OF ARC) INCHES	
		PER FOOT	IN ANY THREE FEET
ALL SIZES	OVER 3% OF OD BUT NOT OVER .50 INCH	.010	.030

(2) QUENCHED AND TEMPERED - THE DEPARTURE FROM STRAIGHTNESS OF TUBING 5 INCHES OR LESS NOMINAL OD SHALL NOT EXCEED .060 INCHES IN ANY 3 FEET OF LENGTH OR .020 INCHES TIMES THE LENGTH IN FEET FOR SHORTER LENGTHS.

LENGTH: LENGTH SHALL VARY NOT MORE THAN +1/8; -0 INCHES.

FOR STANDARD SIZES FOR ROUND TUBING SEE DRAWING AND10102.

Ⓐ CANCELED AFTER 5 MAY 1967. USE AMS2253D.

APPROVED 28 Apr 1953 REVISED Ⓐ 5 May 1967

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33529(ASG)
Navy-Buaer Air Force	A - N - AF -	TOLERANCES - SEAMLESS STEEL TUBING, AIRCRAFT AIRFRAME, ROUND	
PROCUREMENT SPECIFICATION		SUPERSEDES: AND10368	SHEET 1 OF 1

DD FORM 672-1
1 OCT 52

*U.S. GOVERNMENT PRINTING OFFICE 1967-251-518/6248

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